Applied Data Analysis (CS401)

Lecture 3
Visualizing data
24 Sep 2025

EPFL

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Announcements

- You must find 4 team mates and register by **this Fri Sep 26th 23:59**
  - Every student must register individually
  - Still looking for a team or for team members? Use Ed! If need be, talk to Sevda & Pierre [TAs]
- Project milestone 1 was released last Sun; **due Fri 3 Oct 23:59**
- Friday's lab session:
  - 15:30–16:30: Project milestone P1 office hours (Zoom)
  - In parallel: Exercise on data visualization and working with data from the Web (Exercise 2)
  - Happening **only in GCC330, CE14 (not ~~BCH2201~~!)**

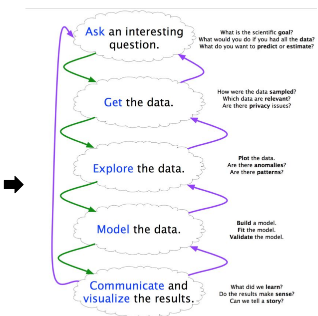
Feedback

Give us feedback on this lecture here:
https://go.epfl.ch/ada2025-lec3-feedback

- What did you (not) like about this lecture?
- What was (not) well explained?
- On what would you like more (fewer) details?
- …

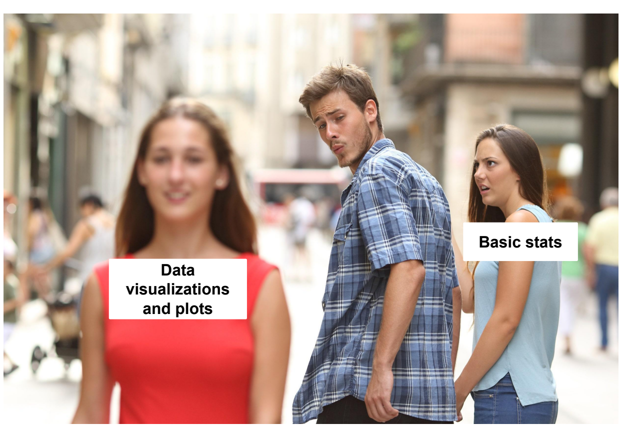
Uses for data visualization

Analysis: Support reasoning about information
- Finding relationships
- Discover structure
- Quantifying values and influences
- Should be part of the data analysis cycle ➡



Communication: Inform and persuade others
- Capture attention, engage
- Tell a story visually
- Can focus on certain aspects and omit others

Decision making: Make it easier to evaluate potential courses of action



An unconventional example

"Garden of Eden": 8 lettuces, each of which is enclosed in its own airtight plexiglas box and represents a major city. The concentration of ozone in each box is controlled in real-time to reflect the current pollution level in the city.

Static viz

Great for data exploration, developed throughout the last few centuries…

Interactive viz

More and more common when delivering the results (and also during exploration). New frameworks are the key enabler.

Want to learn more?

Dedicated course:

[COM-480: Data Visualization]

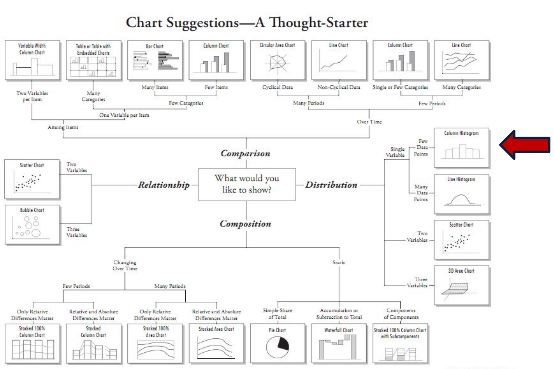
Today's lecture

- Part 1: Navigating the chart landscape
- Part 2: Principles and best practices
- Part 3: A (small) selection of use cases for data visualization

Part 1

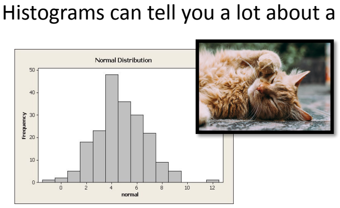
Navigating the chart landscape

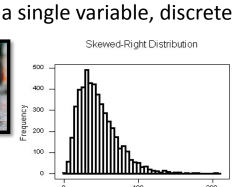
Chart selection



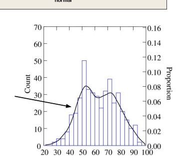
One variable: histograms

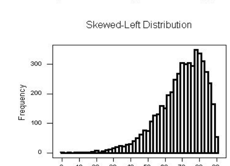
Histograms can tell you a lot about a single variable, discrete or continuous





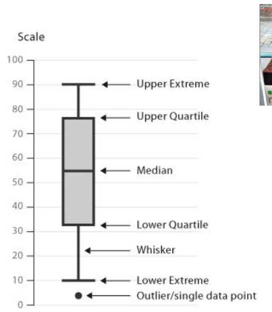
Smoothed histogram (a.k.a. kernel density estimate)

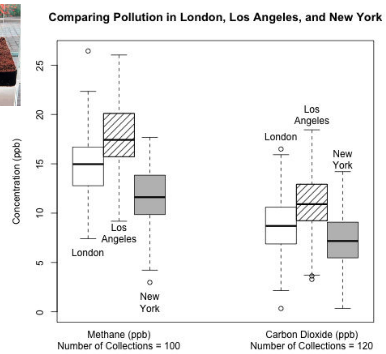




Easy to recognize skewed distributions!

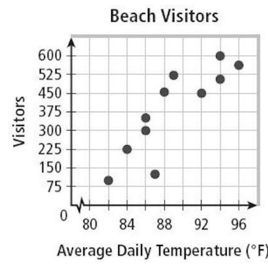
One variable: box plots





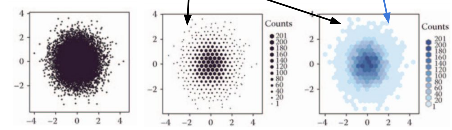
Two variables: scatter plots

Scatter plots quickly expose the relationships between two variables



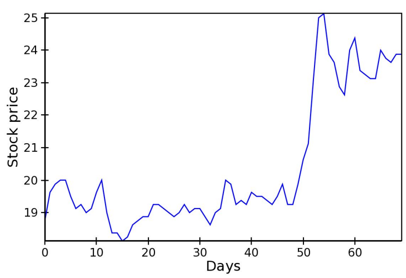
2D histograms

a.k.a. heatmap

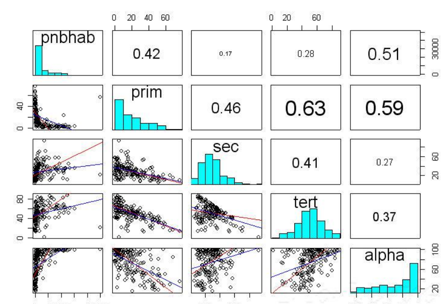


Two variables: line plots

If relationship is functional (for instance, after binning and aggregating)



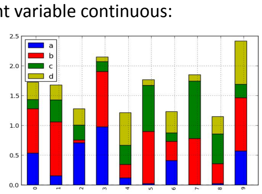
> 2 variables: scatter plot matrix



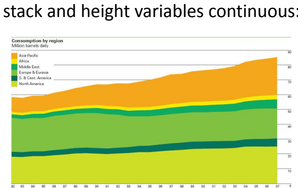
> 2 variables: stacked plots

Here: 3 variables: stack index, height, color

Stack variable and color variables categorical, height variable continuous:

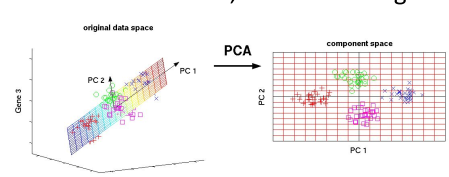


Color variable categorical, stack and height variables continuous:

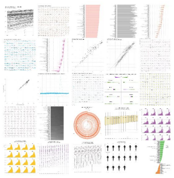


Dimensionality reduction

- For example, **PCA:** allows visualization of high-dimensional continuous data in 2D using principal components
- The principal components are the strongest (highest variation) dimensions in the dataset, and are orthogonal



One dataset, visualized 25 ways



http://flowingdata.com/2017/01/24/one-dataset-visualized-25-ways

"You must help the data focus and get to the point. Otherwise, it just ends up rambling about what it had for breakfast this morning and how the coffee wasn't hot enough."

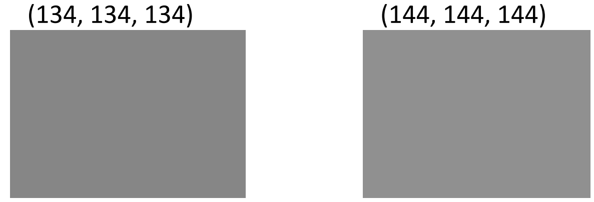
Part 2
Principles and best practices

Instructive coffee table books by Edward Tufte

[another great coffee table book]

Perception of magnitudes

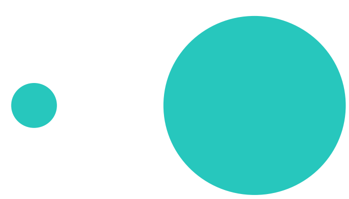
(134, 134, 134) (144, 144, 144)



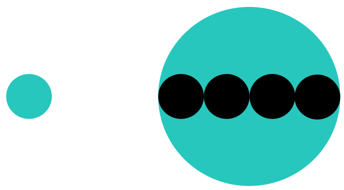
Which is brighter?

Just noticeable difference (JND)

- Weber's law: $$\frac{\Delta I}{I} = k,$$
- I: intensity; ΔI: increase from I to notice a difference; k: constant
- Required increase ΔI depends on original intensity I
- Most continuous variations in stimuli are perceived in discrete (multiplicative) steps



Compare area of circles

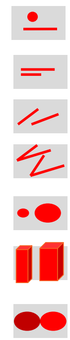


Compare area of circles

Perception of magnitudes

Most accurate

Least accurate



Position

Length

Slope

Angle

Area

Volume

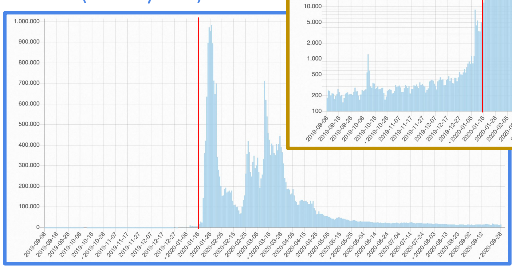
Color hue-saturation-density

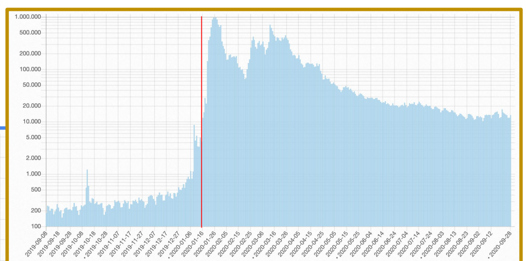
Cleveland & McGill (1984) *Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods*

Choose your axes wisely!

Time series of pageviews of Wikipedia article about “Coronavirus”

(linear y-axis)

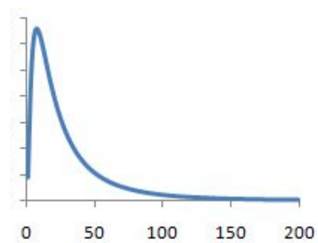




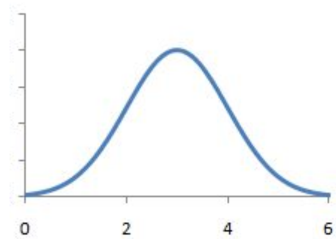
(logarithmic y-axis)

[link]

Choose your axes wisely: Visualizing heavy-tailed distributions



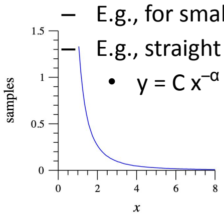
Linear x-axis

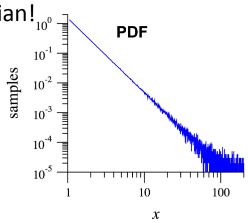


Logarithmic x-axis

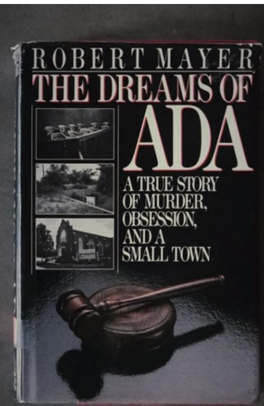
Heavy-tailed data: power laws

- Probability of x: $p(x) = Cx^{-\alpha}$,
  - Very large values are rare, “but not very rare”
  - Body size vs. city size
  - Many natural phenomena are power laws (e.g., # of friends)
  - For dealing with them, need to know some tricks
  - E.g., for small α, mean & var = ∞ → use median!
  - E.g., straight line on log-log axes:
    - $y = C x^{-\alpha}$ ↔ <u>log(y)</u> = <u>log(C)</u> – α <u>log(x)</u>





Commercial break



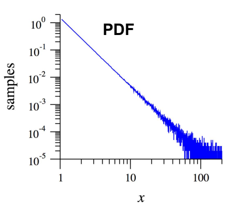
Heavy-tailed data: power laws

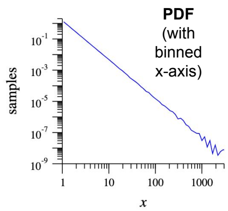
- Complementary cumulative distribution function (CCDF):

P(x) := Pr{X >= x}

POLLING TIME

- Do you recall how is CDF defined?
- Scan QR code or go to https://go.epfl.ch/ada2025-lec3-poll



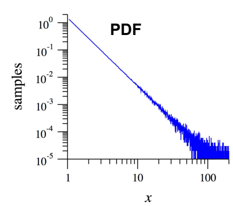


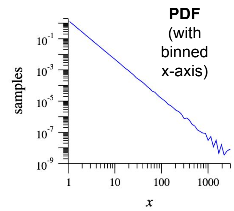
Heavy-tailed data: power laws

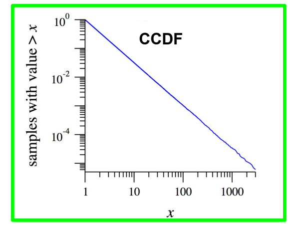
- Complementary cumulative distribution function (CCDF): P(x) := Pr{X >= x}
- CCDF of power law is also a power law (with exponent α – 1)

$$P(x) = C \int_x^\infty x'^{-\alpha}\, \mathrm{d}x' = \frac{C}{\alpha - 1} x^{-(\alpha-1)}.$$

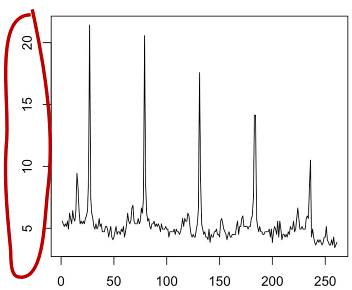
- CCDF plot is monotonically decreasing (even without binning)

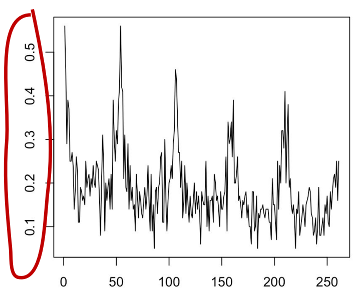






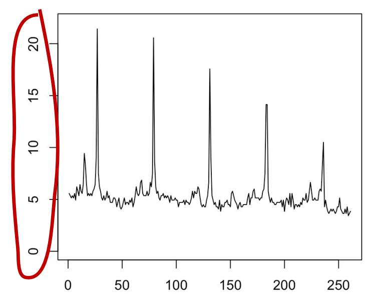
Answer fast: which time series has a higher mean value?

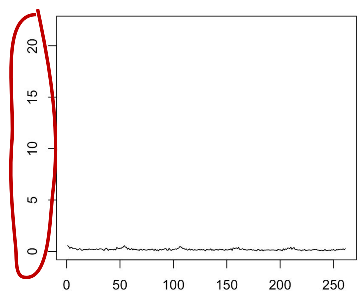




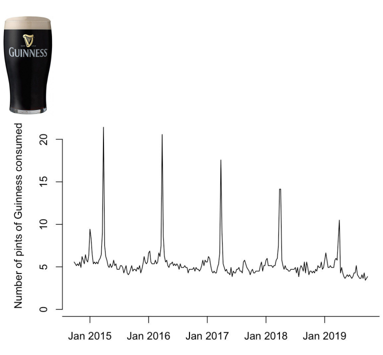
Use consistent axes!

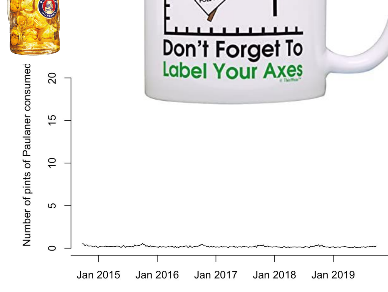
Answer fast: which time series has a higher mean value?

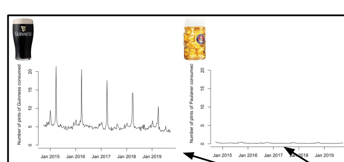




Label your axes!



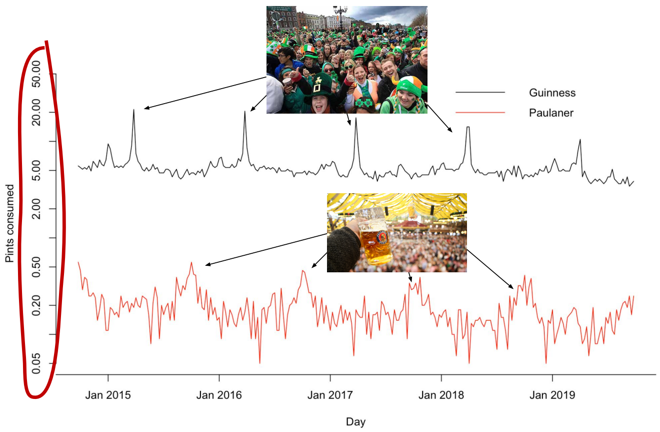




THINK FOR A MINUTE:

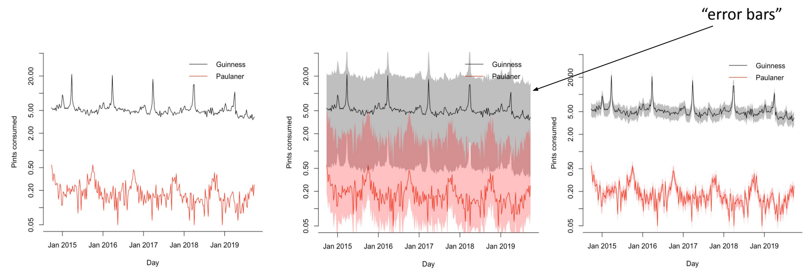
How could we show details of both time series without using different y-axes?

(Feel free to discuss with your neighbor.)

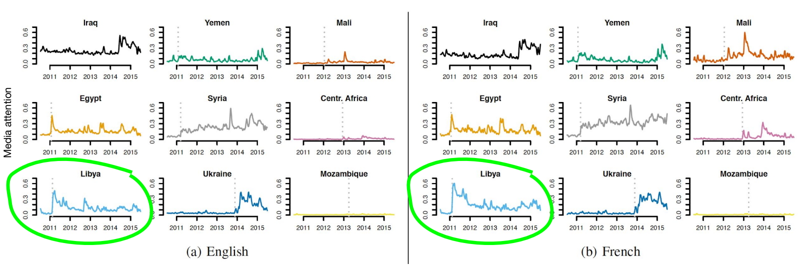


Show data uncertainty!

Which beer is more popular, Guinness or Paulaner?



Consider using small multiples!



(a) English

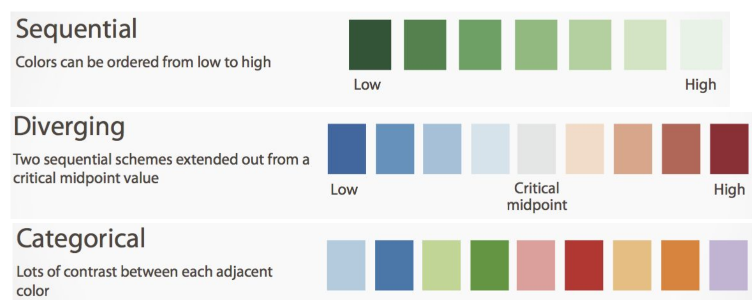
(b) French

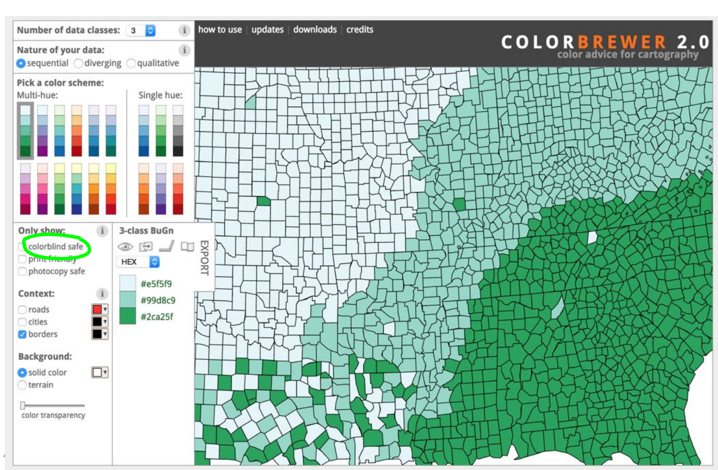
Use colors consistently!

[link]

Use colors wisely!

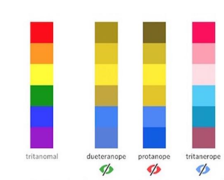
Choose colors based on the information you want to convey

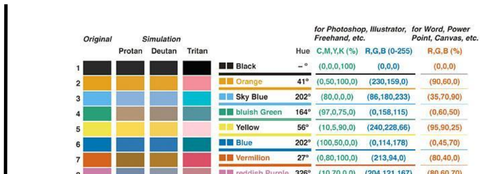




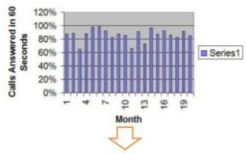
Use colorblind-safe palettes!

- Remember: 10% of males have some form of colorblindness

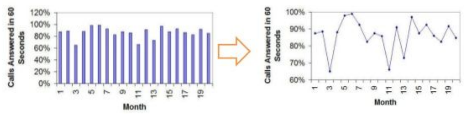




Use data ink wisely! Avoid chart junk!

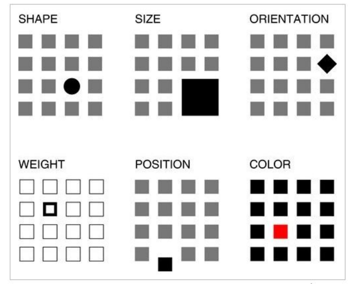


Graphical excellence gives the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space.
-Edward Tufte

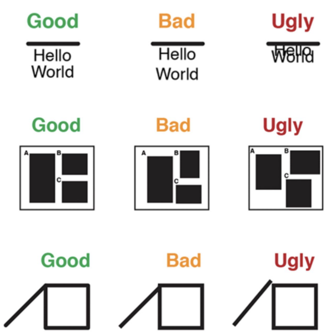


Use visual contrast

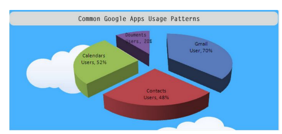
- Use visual contrast to highlight the important information
- Vary the size, shape, position, orientation, or color of an element to make the key part of the figure the most visually prominent
- Avoid incorporating every form of visual contrast

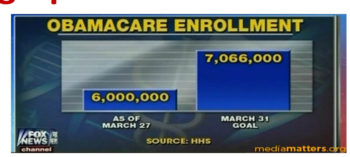


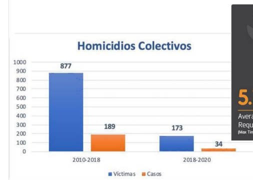
The good, the bad and the ugly

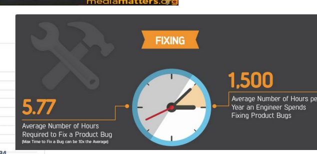


Which principles and best practices do these graphics violate?









Courtesy of [viz.wtf](viz.wtf)

Part 3

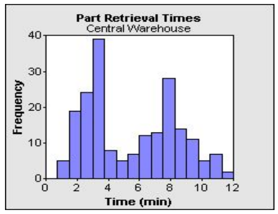
A (small) selection of use cases for data visualization

Use case: Presenting scientific results

Use case:
Data wrangling

Multimodal data

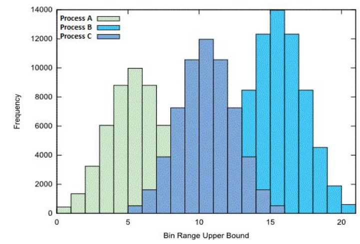
- Two or more distinct peaks in a histogram often **suggest 2 or more distinct populations** of samples.
- But don't guess! Explore further by using, e.g., color and a histogram of multiple populations (p.t.o.).



Use case:
Data wrangling

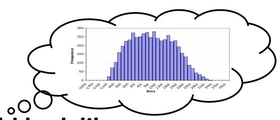
Multimodal data

Explore further by using, e.g., color and a histogram of multiple populations

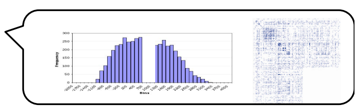


Use case:
Data wrangling

Weird data



- Maintain a **theory of what the data should look like**.
- Some data is **very hard to explain**.
- Never just blink it away!
- First, assume a bug. Try to fix it.
- If not a bug: you might have made an interesting discovery!
- Some of science's most important findings were made by not ignoring weird data, but dwelling on it!



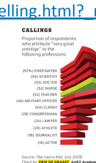
[link]

Use case:
Journalism

NY Times interactive visualizations (recession/recovery 2014)
http://www.nytimes.com/interactive/2014/06/05/upshot/how-the-recession-reshaped-the-economy-in-255-charts.html

And 2014 "the year in interactive storytelling"
http://www.nytimes.com/interactive/2014/12/29/us/year-in-interactive-storytelling.html?_r=0

NY Times graphics are a great source of best practices in viz (except for when they're not…)

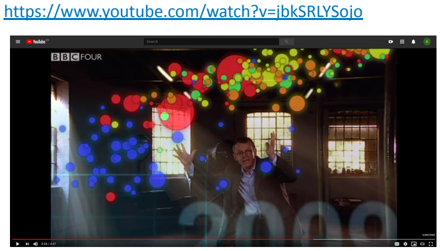


Use case:
Educating the public

Hans Rosling:

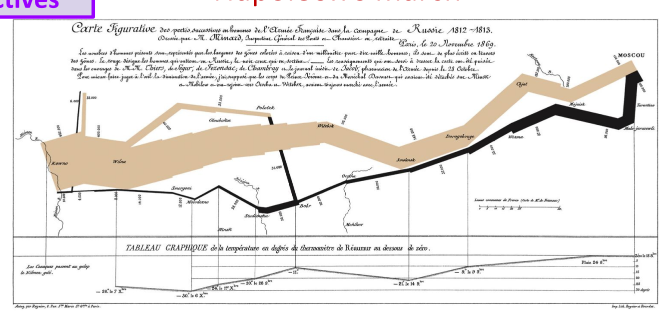
200 countries, 200 years, 4 minutes

https://www.youtube.com/watch?v=jbkSRLYSojo



Use case:
Give new perspectives

Charles Joseph Minard 1869
Napoleon's march



According to Tufte: "*It may well be the best statistical graphic ever drawn.*"
5 variables: army size, location, dates, direction, temperature during retreat

Tools

(remaining slides for your personal perusal)

Interactive toolkits: D3

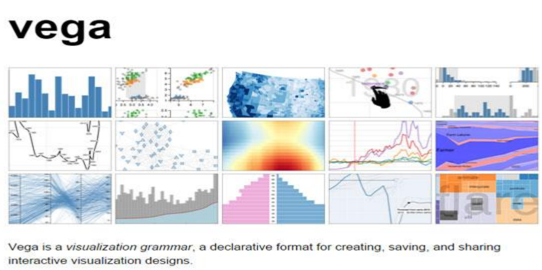
Without doubt, the most widely used interactive visualization framework is **D3**.

Note from the authors: *D3 is intentionally a low-level system. During the early design of D3, we even referred to it as a "visualization kernel" rather than a "toolkit" or "framework"*

Interactive toolkits: Vega

Vega is a "visualization grammar" developed on top of D3.js

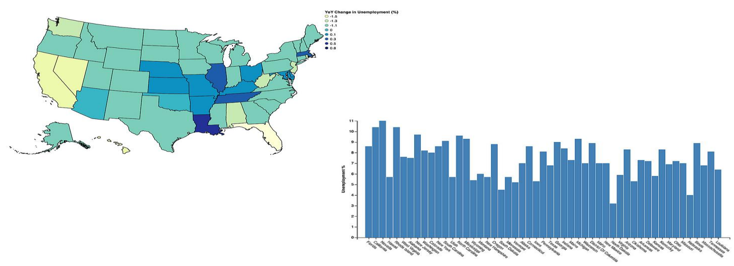
It specifies graphics in JSON format.



Interactive toolkits: Vincent

Vincent is a Python-to-Vega translator.

Trivia question: why is it called Vincent? Hint: Vincent+Vega= ?



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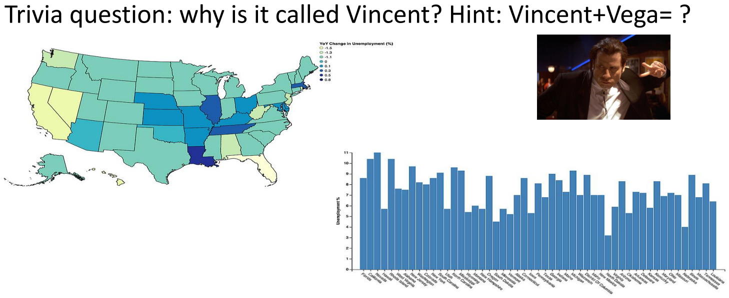
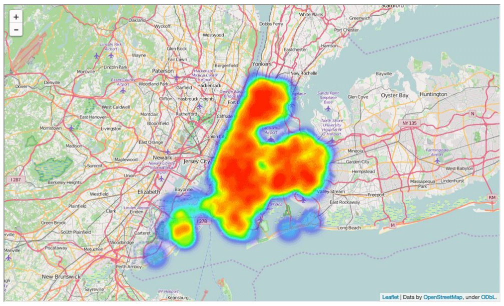


Bokeh: another interactive viz library

Bokeh is an independent Viz library focused more heavily on big data visualization. Has both Python and Scala bindings.

Visualizing maps: Folium

More in tomorrow's lab session!



Feedback

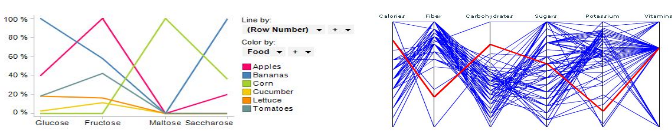
Give us feedback on this lecture here:
https://go.epfl.ch/ada2025-lec3-feedback

- What did you (not) like about this lecture?
- What was (not) well explained?
- On what would you like more (fewer) details?
- …

> 2 variables: parallel-coord. plots

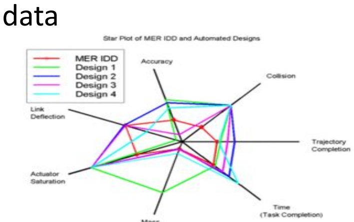
Color, x, y

Color variable is categorical, others arbitrary



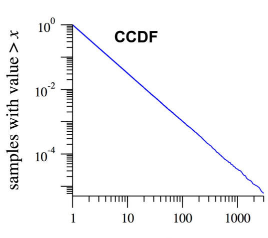
> 2 variables: radar charts

- Similar to parallel-coord. plots
- Doesn't pretend that x axis has meaningful order
- Also good for periodic data



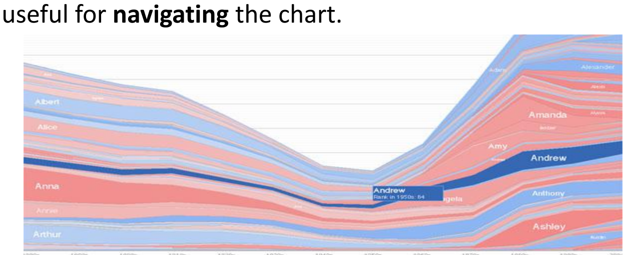
Heavy-tailed data: power laws

- Smart trick for plotting CCDF of any distribution:
  - x-axis: data sorted in ascending order
  - y-axis: (n:1)/n (where n is number of data points)



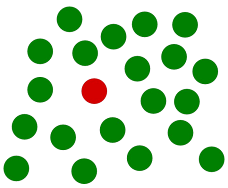
Interactive chart design: simplifying

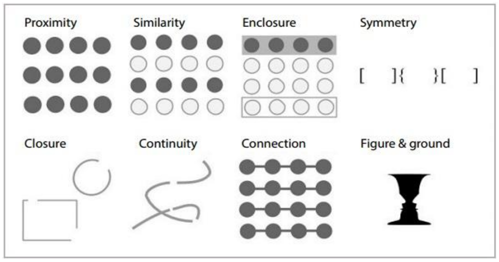
- With interactive charts you can keep things very simple by **hiding** and **dynamically revealing** important structure.
- On an interactive chart, you reveal the information most useful for **navigating** the chart.



Use structure!

Gestalt psychology principles (1912)



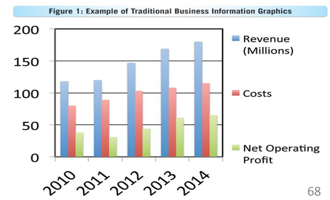


A case for ugly visualizations

People instinctively gravitate to attractive visualizations, and they have a better chance of getting on the cover of a journal.

But does this **conflict with the goals of visualization?**

- Rapid exploration
- Focus on most important details
- Easy and fast to develop and customize



Guide your audience!